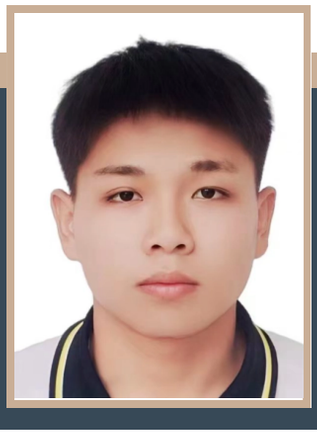

郑恩赐

PERSONAL RESUME

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Personal Information

- Name: Zheng Enci
- Phone: +86 13174662233
- Email: zheng_enci@qq.com
- Gender: Male
- Age: 20
- Location: Xiamen, Fujian Province

Job Intention

- Target Position: AI Full-Stack Application Development Engineer
- Desired Location: Xiamen
- Job Type: Full-time
- Expected Salary: Negotiable

Education

- Education Period: 2023.09 - 2027.06
- Institution: Xiamen Institute of Technology, School of Artificial Intelligence
- Major: Software Engineering (AI Direction)
- Degree: Bachelor
- Education Type: Regular Enrollment
- Education Content:
  - **Core Courses**: Data Structures, Algorithms, Machine Learning, Computer Vision, Web Development.
  - **Academic Focus**: Specialized in AI application development and full-stack engineering, with hands-on project experience in computer vision and system architecture.

Project Experience

- Project Period: 2025.10 - 2025.12
- Project Title: Hongxing Farm Corn Germination Monitoring System (Collaboration with Agricultural Academy)
- Role: Frontend Developer / AI Application Developer
- Project Content: Project Background
  - Commissioned by the Agricultural Academy to develop a YOLO-based corn germination monitoring system for Beidahuang Group, replacing inefficient manual field inspections and enabling timely identification of missing seedling areas.

  Core Contributions
  - **Frontend Visualization**: Independently developed the frontend visualization system using HTML5 Canvas (Fabric.js) to parse YOLO annotation data, implementing seedling point grid aggregation and heatmap generation.
  - **Parameter Adaptation**: Developed multi-parameter adjustment features for grid size, rotation, and translation to adapt to irregular field ridges, **reducing row spacing error to within 5 pixels**.
  - **Workflow Closure**: Designed and implemented a manual correction panel, forming an "algorithm detection + manual review" closed-loop workflow with integrated germination rate calculation.

  Project Outcomes
  - **Compressed the full process from detection to heatmap generation to manual review from several hours to under 10 minutes, significantly enhancing field inspection efficiency**.
  - Project open-sourced on Gitee/GitHub, providing a reusable technical solution for agricultural intelligence.

- Project Period: 2025.09 - 2025.12
- Project Title: Student Management System
- Role: Full-Stack Developer
- Project Content: Project Background
  - Developed an integrated student information and attendance management system for Xiamen Institute of Technology's AI Creation Workshop, addressing issues of irregular information management, unrestricted sign-in times, and lack of data visualization.

  Core Contributions
  - **Architecture Design**: Independently responsible for frontend and backend architecture design and implementation, utilizing Vue3 + Spring Boot 3.5.5 + MySQL stack to complete core functionality development.
  - **Security Mechanism**: Designed JWT authentication and IP whitelist dual security mechanisms, integrated CAPTCHA verification, **completely eliminating sign-in fraud with 100% security**.
  - **Performance Optimization**: Introduced Redis caching to optimize high-frequency queries, **reducing sign-in status response time from over 500ms to under 100ms, achieving an 80% performance improvement**.
  - **Loading Strategy**: Implemented on-demand component library loading and parallel data loading strategies, **significantly enhancing first-screen load speed and real-time data dashboard update latency**.

  Project Outcomes

- System successfully deployed and running stably, supporting multi-user concurrent access with daily sign-in requests exceeding one thousand.
- Project open-sourced on GitHub and Gitee, receiving widespread use and positive feedback within the institution.

- Project Period: 2025.06 - 2025.07
- Project Title: Chinese Academy of Sciences Urban Environment Institute Aerial Camera Image Stitching System
- Role: Deep Learning Algorithm Engineer
- Project Content: Project Background
  - Undertook a project for the Chinese Academy of Sciences Urban Environment Institute to develop a deep learning-based multi-channel aerial camera video stream image stitching system, addressing traditional method challenges in complex scenes.

  Core Contributions
  - **Network Architecture**: Designed and implemented an end-to-end deep learning network architecture using dual-branch CNN regression networks to predict homography and TPS transformation parameters.
  - **Feature Extraction**: Utilized pre-trained ResNet50 for multi-scale feature extraction, enabling CCL feature matching and feature flow computation for robust transformation prediction input.
  - **Algorithm Implementation**: Independently developed DLT, TPS, and other image transformation algorithms, designing a multi-level loss function including L1 loss, color consistency, and grid constraints.
  - **Performance Optimization**: Employed CUDA for GPU acceleration, optimizing the entire pipeline from image preprocessing to transformation and fusion.

  Project Outcomes
  - **The developed deep learning stitching method significantly outperformed traditional algorithms in complex scenes, meeting high-precision urban environmental monitoring requirements**.
  - Completed the entire process from requirement对接, algorithm R&D to delivery validation, receiving high recognition from the institute's technical team.

- Project Period: 2025.03 - 2025.04
- Project Title: Classroom Anomaly Behavior Detection System (EduBot)
- Role: AI Algorithm Engineer / Full-Stack Developer
- Project Content: Project Background
  - Participated in the university's AI education reform project to develop an intelligent system for real-time detection of classroom anomalies such as desk resting, phone use, standing, and looking up.

  Core Contributions
  - **Data Construction**: Used LabelImg to independently build an annotated dataset and trained a specialized detection model based on PyTorch and YOLOv8.
  - **Architecture Innovation**: Innovatively designed a two-stage architecture of "YOLO for human detection + proprietary behavior classification model", **achieving significantly higher model accuracy than general open-source models**.

- **Full-Stack Development**: Independently completed full-stack development and integration of the system's frontend (display and interaction), backend (Flask API), and model service components.
- **Function Expansion**: Integrated large language models like Tongyi Qianwen to expand AI-assisted teaching features such as PPTX-to-mind map conversion and intelligent lesson plan generation.

Project Outcomes

- Project awarded **National College Student Innovation and Entrepreneurship Training Program 立项 (Project Number: 202513115006)**.
- Filed invention patent **"Multi-Modal Large Model-Based Classroom Behavior Detection Method" as third inventor (Application Number: CN202510645941.9)**.
- Project frontend open-sourced on GitHub and Gitee, providing technical reference for educational intelligence.

Internship Experience

- Internship Period: 2024.07 - 2024.08
- Company: Xiamen Hongchen Network Technology Co., Ltd.
- Industry: Internet/Software Development
- Position: Full-Stack Developer Intern
- Internship Content:
  - **Full-Stack Development**: Independently completed full-stack development of an enterprise-level human resource management information system, covering 7 core business modules including employee management, attendance clock-in, and department management.
  - **Permission System**: Designed and implemented a JWT Token-based authentication and authorization mechanism, establishing a dual-role permission system for administrators and employees.
  - **Technical Practice**: Utilized Vue3 + Spring Boot + MyBatis technology stack, gaining deep understanding of frontend-backend separation architecture and RESTful API design standards.
  - **Project Delivery**: Completed the entire process from requirement analysis, technology selection, development implementation to deployment and launch, successfully delivering a stable running system.

Skills

- **Frontend Development**: Vue3 (Composition API), Vue Router, Pinia, Element Plus, ECharts, Fabric.js, TypeScript, Axios, Vite, WebSocket.
- **Backend Development**: Spring Boot 3.x, Spring MVC, Spring Security, JWT, Spring Data JPA, MyBatis, Flask, SQLAlchemy, RESTful API, Java 21/24.
- **AI/Machine Learning**: PyTorch, YOLOv5/v8, OpenCV, Object Detection, Behavior Recognition, Model Training & Deployment, Image Stitching, LangChain, OpenAI API, CUDA Acceleration.
- **Large Language Models**: Application development with Tongyi Qianwen, Zidong Taichu, and other multi-modal large models.
- **Database**: MySQL, PostgreSQL, Redis, SQL Optimization, Index Design, Caching Strategies.

- **Tools & Deployment**: Git, Docker, Nginx, Maven, Linux, Postman, WebStorm, Cursor, Intranet Penetration, Project Deployment.

Awards & Honors

- **2025.09**: National College Student Innovation and Entrepreneurship Training Program立项 (Project Number: 202513115006) - "EduBot – Creator of New Educational Forms".
- **2025.09**: Invention Patent (Second Inventor) - "A Student Behavior Detection System and Method Based on Classroom Monitoring" (Application Number: CN202511244232.6).
- **2025.05**: Invention Patent (Third Inventor) - "Multi-Modal Large Model-Based Classroom Behavior Detection Method" (Application Number: CN202510645941.9).

Self-Assessment

Enthusiastic about exploring and practicing artificial intelligence and full-stack development technologies, with excellent independent problem-solving abilities and innovative thinking. Capable of quickly learning and applying new technologies, successfully integrating AI algorithms with engineering practices in multiple projects, yielding substantive outcomes including national-level project approvals and invention patents. Possesses strong full project lifecycle management skills, with rich experience from requirement analysis, technology selection, development implementation to deployment and maintenance. Code-standardized, focused on performance optimization and user experience, with strong teamwork spirit and resilience under pressure.